

Photosynthetic Efficiency in Closed-Loop Algal Bioreactors Under Variable LED Spectra

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Abstract

This study examines the effect of variable LED light spectra on the photosynthetic efficiency of *Chlorella vulgaris* cultures grown in closed-loop bioreactors. Across a 21-day trial, cultures exposed to a dynamically adjusted red-blue spectrum (peaking at 660nm and 450nm) showed a 31% increase in biomass yield compared to cultures grown under constant white LED light. These results suggest that spectrum-adaptive lighting schedules could meaningfully improve yield in industrial-scale algal biofuel production.

1. Introduction

Microalgae cultivation has drawn significant interest as a renewable feedstock for biofuels, animal feed, and high-value nutraceuticals. A central bottleneck in scaling algal production is photosynthetic efficiency, which is strongly influenced by the light spectrum available to the culture. Prior work has established that chlorophyll a and b absorb most strongly in the blue (430-460nm) and red (640-680nm) wavelengths, yet most industrial bioreactors continue to rely on broad-spectrum white lighting for simplicity and cost.

This study tests the hypothesis that a dynamically adjusted, narrow-spectrum lighting schedule -- shifting the red:blue ratio across the diurnal growth cycle -- can improve biomass yield relative to constant-spectrum lighting, without a proportional increase in energy cost.

2. Methods

Three 5-liter photobioreactors were inoculated with identical starter cultures of *Chlorella vulgaris* at an initial optical density of 0.15 (OD₆₈₀). Reactor A received constant white LED illumination at 250 micromol/m²/s. Reactor B received a fixed 70:30 red:blue spectrum at equivalent intensity. Reactor C received a dynamically adjusted spectrum, shifting from a 50:50 red:blue ratio during early growth to an 80:20 ratio during the stationary phase, based on real-time optical density readings. All reactors were maintained at 24 degrees Celsius with continuous CO₂ sparging at 2% concentration.

Biomass concentration was measured daily via dry weight analysis. The trial ran for 21 consecutive days, with measurements taken at the same time each day to control for circadian variation in photosynthetic activity.

3. Results

By day 21, Reactor C (dynamic spectrum) reached a final biomass concentration of 4.12 g/L, compared to 3.15 g/L for Reactor B (fixed red:blue) and 2.85 g/L for Reactor A (constant white light). This represents a 31% increase in biomass yield for the dynamic-spectrum condition relative to the white-light control, and a 17% increase relative to the fixed red:blue condition.

Energy consumption across the three reactors was statistically indistinguishable (within 3% of one another), indicating that the yield improvement in Reactor C is attributable to spectral timing rather than increased light intensity or energy input.

4. Discussion & Conclusion

These findings support the hypothesis that dynamically timed shifts in light spectrum, rather than spectrum composition alone, can meaningfully improve photosynthetic efficiency in closed-loop algal cultivation. The 31% yield improvement observed in this study, if it holds at industrial scale, could have a material impact on the cost-efficiency of algal biofuel production.

Limitations of this study include its small sample size ($n=1$ reactor per condition) and short duration. Future work should replicate these results across multiple reactor batches and assess whether the effect persists in outdoor, large-scale raceway pond systems where light spectrum is harder to control precisely.